

Library Management System Python Code

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class Library:
    def __init__(self):
        self.books = []
        self.patrons = []
        self.borrowed = []
    # Add a book
    def add_book(self, book):
        self.books.append(book)
        print(book, "added successfully.")
    # Register a patron
    def register_patron(self, patron):
        self.patrons.append(patron)
        print(patron, "registered successfully.")
    # Borrow a book
    def borrow_book(self, name, book):
        if name in self.patrons and book in self.books:
            self.books.remove(book)
            self.borrowed.append([name, book])
            print(name, "borrowed", book)
        else:
            print("Book or Patron not found.")
    # Return a book
    def return_book(self, name, book):
        if [name, book] in self.borrowed:
            self.borrowed.remove([name, book])
            self.books.append(book)
            print(name, "returned", book)
        else:
            print("Record not found.")
    # Display available books
    def display_books(self):
        if len(self.books) == 0:
            print("No books available.")
        else:
            print("\n\nAvailable Books:")
            for book in self.books:
                print(book)
SOC_library = Library()
SOC_library.add_book("Python")
SOC_library.add_book("Java")
SOC_library.add_book("C++")
SOC_library.register_patron("Raj")
SOC_library.register_patron("Padma")
SOC_library.display_books()
SOC_library.borrow_book("Raj", "Python")
SOC_library.borrow_book("Padma", "Python")
SOC_library.display_books()
SOC_library.return_book("Raj", "Python")
SOC_library.display_books()
```